



Society of Petroleum Engineers

SPE-227161-MS

Analyses, Challenges and Solutions for the Electrification of Rigless Well Operations in the Arabian Gulf Region

Ayman. A. Alomar, Saudi Aramco, Dhahran, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227161-MS](https://doi.org/10.2118/227161-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper identifies challenges and offers solutions for deploying electricity-powered equipment in rigless and well completion operations in the Arabian Gulf region. It aims to replace diesel-powered systems, reducing Scope 1 and 3 greenhouse gas emissions and mitigating environmental impact through the adoption of more sustainable energy sources for oil and gas operations.

This study analyzes commercially available equipment and deployment challenges, as well as power and fuel systems in the Arabian Gulf. It examines regional regulations, logistics, and installation requirements. A comprehensive review of international best practices in the US and China informs the research, which draws on online resources, industry reports, and site visits to oil and gas fields and operations in these regions, adapting global insights to the local context and identifying opportunities for improvement.

The paper's analysis yielded seven primary electricity-powered equipment and their corresponding power requirements, namely Slickline, Eline, coiled tubing, fluid pump, HP frac pump, sand conveyor, and site facilities including accommodations. The research comprehensively identified all available power and fuel options in the region, determining the most suitable choices for each application and location. These options encompassed the accessibility of grid power, requisite equipment, and procedures for connection, as well as portable energy storage systems and diverse types of fuel for operating field gas turbines, including liquefied natural gas (LNG), compressed natural gas (CNG), liquefied petroleum gas (LPG), and wellhead gas. Each of these power and fuel alternatives presented unique challenges pertinent to specific areas, such as regulatory hurdles for grid connectivity, fuel gas transportation, or treatment of wellhead gases. Ultimately, the paper concludes that the rapid deployment of electricity-powered equipment in Arabian Gulf fields is feasible, enabling the reduction or elimination of Scope 1 and 3 greenhouse gas emissions and facilitating an energy transition; nonetheless, regional collaboration is essential to overcome these challenges and devise cost-effective solutions.

This paper presents novel findings based on a 3-year comprehensive analysis of the region. It incorporates insights from US and China field visits, manufacturer engagements, and attempted deployments of electricity-powered Eline units in the region. The paper considered integrated systems to operate these equipment using different power sources and fuels to overcome specific areas challenges and power availabilities.

Introduction

The oil and gas industry is shifting towards more sustainable and environmentally friendly operations, with a focus on reducing greenhouse gas emissions. In the Arabian Gulf region, there is a growing interest in adopting electricity-powered equipment for rigless well operations, aiming to replace traditional diesel-powered systems and minimize environmental impact. This paper explores the challenges and solutions for deploying electricity-powered equipment in the region, analyzing available equipment, power and fuel systems, and regional regulations with comparison of practices in China and United States.

Well Intervention Electricity-Powered Equipment

The increasing demand for efficient and environmentally friendly well intervention operations has led to the development and adoption of electricity-powered equipment. These systems offer a cleaner, quieter, and more reliable alternative to traditional diesel-powered equipment, reducing the environmental impact and improving the overall safety of well intervention operations.

Electricity-powered well intervention equipment includes a range of tools and systems, such as electric slickline, E-line, coiled tubing units, and hydraulic fracturing pumps. These systems are designed to operate using electrical power, which can be supplied from a variety of sources, including the electrical grid, generators, or mobile energy storage systems.

Table 1—Table shows estimations and scenarios of electrical power requirements for most common types of well intervention electricity-powered equipment operations.

OPERATION /EQUIPMENT	4xE- FRAC PUMP	E-BATCH MIXER	E-SAND BLENDER	E- DATA VAN	E-IH & E- REEL	E- WINCH	TOTAL POWER REQUIREMENT
PROP FRAC	15 MW		115 KW	15 KW			15.13 MW
ACID FRAC	15 MW	115 KW		15 KW			15.13 MW
COILED TUBING		115 KW			250 KW		0.365 MW
SL/EL						150 KW	0.150 MW

The adoption of electricity-powered well intervention equipment is expected to continue to grow as the industry seeks to reduce its environmental footprint and improve the efficiency and safety of well intervention operations. As the technology continues to evolve, we can expect to see further innovations and improvements in the design and operation of electricity-powered well intervention equipment.

The adoption of electricity-powered well intervention equipment is gaining momentum in the Arabian Gulf region, but it is hindered by challenges in providing reliable electrical energy sources at well sites. Several factors contribute to these challenges, which will be discussed in the following sections.

Electrical Energy Sources

The oil and gas industry is exploring alternative power sources for well intervention equipment to reduce greenhouse gas emissions and increase efficiency. Electrical power from the grid, diesel power generators, mobile gas turbine gensets, and mobile energy storage systems are being considered as viable options.

Electrical Power Grid

The utilization of electrical power from the grid represents the most viable option for operating electrical well intervention equipment, offering a dual benefit of reduced greenhouse gas (GHG) emissions and a reliable energy source. The electrical power grid can provide a sustainable power supply with significantly lower emissions, which are categorized under Scope 2. Notably, the majority of power plants in the Arabian

Gulf region that feed the electrical grids rely on natural gas for power generation, further enhancing the environmental benefits.

In terms of electrical grid availability at well sites, the Arabian Gulf region's fields are generally well-covered due to the necessity of remote well monitoring, flow control, and the operation of artificial lift pumps, scale, and corrosion inhibitor injection systems. However, the mere availability of the electrical grid is insufficient to support the temporary operation of electrical well intervention equipment, which demands specific power parameters, including high voltage (320–480 V) and high power capacity, ranging from 150 kVA for E-line operations to 15,000 kVA for major fracturing operations.

To overcome this challenge, countries such as China and the United States have implemented innovative solutions. For instance, in cases where higher voltage and moderate power capacity are required for small pumping operations, portable electrical transformers are employed. Furthermore, in China, temporary electrical substations are installed to support major fracturing operations when economically feasible. These approaches demonstrate the potential for the oil and gas industry to adopt sustainable and efficient power supply solutions for well intervention equipment, minimizing environmental impact while ensuring operational efficiency.

Diesel Power Generators

Diesel power generators represent an alternative means of operating electrical well intervention equipment. However, this option is less desirable from an environmental and economic standpoint. The utilization of diesel power generators fails to yield any reductions in greenhouse gas (GHG) emissions, thereby negating a key benefit of transitioning to electrical power. Furthermore, the operational and maintenance costs associated with internal combustion engines are retained, while the fuel efficiency may be compromised due to the additional energy conversion steps involved. In comparison to diesel-powered well intervention equipment, the energy conversion process from diesel to electricity and then to mechanical energy can result in decreased overall efficiency, rendering diesel power generators a less attractive option for powering electrical well intervention equipment. Notably, diesel power generators are only considered in countries such as China and the United States when logistical constraints render it infeasible to transport diesel-powered well intervention equipment, and electrical well intervention equipment are readily available, highlighting the preference for more efficient and environmentally friendly solutions whenever possible.

Mobile Gas Turbine Gensets

Mobile gas turbine gensets have emerged as a viable option for operating electrical well intervention equipment, particularly for major pumping and fracturing operations that require high power capacity exceeding 2 megawatts. One of the key advantages of mobile gas turbine gensets is their flexibility in utilizing different types of fuel, with Compressed Natural Gas (CNG) being the most preferred option due to its compatibility and favorable emission profile, followed by Liquefied Natural Gas (LNG), Wellhead Natural Gas, and Liquefied Petroleum Gas (LPG) as the least preferred alternative. A more detailed discussion on fuel sources and transportation can be found in the Fuel Source & Transportation section of this paper. In countries such as China and the United States, mobile gas turbine gensets with varying power capacities, including multiple 5.5–6 megawatt units or single large mobile gensets capable of delivering up to 32 megawatts, are commonly employed.

However, the implementation of mobile gas turbine gensets in the Arabian Gulf region poses significant challenges, primarily related to the producer availability and transportation of the required fuel. Additionally, the maximum power output of these gensets is inversely affected by ambient temperature, resulting in a notable decrease in power capacity. For instance, a 32 megawatt genset may only produce 25 megawatts when operating in an air intake temperature of 45°C. Despite these challenges, the feasibility of mobile gas turbine gensets becomes more pronounced when performing continuous major pumping and fracturing operations during the winter season in the Arabian Gulf region. Under these conditions, mobile

gas turbine gensets can achieve emission reductions ranging from 20% to 30% compared to diesel-powered well intervention equipment, making them a more environmentally friendly alternative.

Mobile Energy Storage Systems (MESS)

Mobile Energy Storage Systems (MESS) have emerged as a viable solution for powering electrical well intervention equipment, particularly in applications where low power capacity demand and limited or no availability of electrical power grid are prevalent. These systems leverage advanced battery technologies, such as lithium-ion or lithium iron phosphate (LFP), in conjunction with intelligent power management systems, to provide reliable and portable energy without reliance on connected or generated power supply. MESS can be charged prior to dispatch to the well location and can sustain operations for a duration dependent on the power draw and operational requirements. Furthermore, in some cases, these systems can be recharged using mobile gas turbine gensets, enhancing their versatility and autonomy.

The application of MESS has been successfully demonstrated in various regions, including China, where systems with capacities of up to 4 MWh are utilized to operate electrical E-Line and Coiled Tubing equipment. However, the operational duration of these systems is limited, with full consumption of the energy storage capacity occurring within 3 days of E-Line operation or 1 day of Coiled Tubing operation. This necessitates the deployment of multiple systems to extend operational time, which can compromise economic feasibility due to the high upfront costs of these systems.

In the United States, MESS are primarily employed during the startup phase of electrical frac pumps, where the motor draws extra power before stabilizing to normal power demand. This application highlights the potential of MESS to provide peak shaving capabilities and support the efficient operation of electrical well intervention equipment.

Currently, MESS are not readily available in the Arabian Gulf region; however, their importation, acquisition, and operation are relatively straightforward. Over the long term, MESS are likely to become economically feasible in this region, owing to the relatively low energy prices, which will reduce the costs associated with recharging these systems. Additionally, MESS pose fewer risks compared to gas-operated turbine gensets, making them an attractive alternative for powering electrical well intervention equipment in the Arabian Gulf region. As the industry continues to evolve towards more sustainable and environmentally responsible practices, the adoption of MESS is expected to play a significant role in supporting the efficient and reliable operation of electrical well intervention equipment.

Combination Power Mix

In scenarios where a single electrical power source is insufficient or economically unviable to operate electrical well intervention equipment, service providers have adopted a hybrid approach, leveraging a combination of electrical power sources. This strategy has been successfully implemented in countries such as China, where mobile gas turbine gensets are utilized to power frac pumps, while mobile energy storage systems are employed to operate lower-power demand equipment, such as electrical data frac vans. Similarly, in the United States, a dual-power approach is used, where electrical frac pumps are fed by both the electrical power grid and mobile gas turbine gensets simultaneously, via an electrical distribution panel. This innovative solution effectively addresses the power demand spike during frac pump startups, allowing for a reduction in the number of mobile gas turbine gensets required to maintain continuous pumping operations, while ultimately providing the necessary power and minimizing emissions.

The adoption of a hybrid electrical power mix is particularly feasible in the Arabian Gulf region, given the widespread availability of the electrical power grid. This existing infrastructure can be supplemented by either mobile energy storage systems or a limited number of mobile gas turbine gensets, providing a reliable and efficient means of powering electrical well intervention equipment. By combining these power sources, operators in the Arabian Gulf region can optimize their energy supply, reduce their environmental footprint, and improve the overall efficiency of their well intervention operations. This hybrid approach offers a

pragmatic solution for addressing the complex power demands of electrical well intervention equipment, while also promoting a more sustainable and environmentally responsible operating practice.

Fuel Source & Transportation for Mobile Gas Turbine Gensets

Mobile gas turbine gensets require a reliable fuel source, with options including wellhead natural gas, liquefied natural gas (LNG), and compressed natural gas (CNG), each with its own advantages and challenges.

Wellhead Natural Gas

Wellhead natural gas presents a viable option for fueling mobile gas turbine gensets, wherein gas is sourced from a nearby producing well via a temporary pipeline. This method has gained widespread acceptance in the United States, particularly in areas where other types of gases are not readily available or logistically feasible, or when the nearest gas well producer exhibits relatively low sourness conditions. However, the utilization of wellhead natural gas as a fuel source for mobile gas turbine gensets is not without its challenges. The primary obstacle lies in the gas conditioning process, which necessitates a series of complex treatments before the gas can be safely injected into the turbine. These processes typically include a 3/4-phase separator, gas dehydration unit, and sour gas treatment, among others.

The use of wellhead natural gas as a fuel source for mobile gas turbine gensets is particularly problematic in the Arabian Gulf region, owing to the prevalent sour conditions of the wells. The high levels of hydrogen sulfide and other corrosive compounds present in the gas necessitate more stringent and elaborate gas conditioning processes, which can be technically and economically challenging. Furthermore, regulatory approvals to utilize gas well producers for generating energy outside of dedicated power plants can be difficult to obtain, adding an additional layer of complexity to the implementation of this fuel source. As a result, the use of wellhead natural gas to fuel mobile gas turbine gensets in the Arabian Gulf region is often hindered by these technical, logistical, and regulatory hurdles, making it a less attractive option compared to other fuel sources.

Liquefied Natural Gas (LNG)

Liquefied Natural Gas (LNG) is a highly viable option for fueling mobile gas turbine gensets, owing to its widespread popularity and availability, particularly in countries such as China and the United States. The use of LNG as a fuel source requires a gas tank transporter from an LNG plant to deliver the fuel to the well site, as well as a mobile evaporation unit to convert the liquid fuel into a gaseous state, rendering it suitable for consumption by the mobile gas turbine gensets. Despite the additional infrastructure requirements, LNG is an attractive fuel option due to its high energy density in liquid form, which enables a single transporter to provide a substantial amount of gas to operate the turbine, compared to other fuel sources such as Compressed Natural Gas (CNG).

The Arabian Gulf region presents a favorable environment for the adoption of LNG as a fuel source, given its availability in several countries. However, to fully leverage the potential of LNG, it is essential to upgrade the existing transportation regulations in the region to align with the growing demand for operating electricity-powered equipment in well intervention operations. The development of a more comprehensive and supportive regulatory framework would facilitate the safe and efficient transportation of LNG, ultimately enabling the widespread adoption of this fuel source and promoting a more sustainable and environmentally friendly approach to well intervention operations. By addressing the regulatory challenges and investing in the necessary infrastructure, the Arabian Gulf region can capitalize on the benefits of LNG as a fuel source, including reduced emissions and improved energy efficiency, and position itself at the forefront of the industry's transition towards more environmentally responsible practices.

Compressed Natural Gas (CNG)

Compressed Natural Gas (CNG) is considered one of the most viable solutions for fueling mobile gas turbine gensets, owing to its ability to be supplied directly to the turbine genset without requiring evaporating units or prior treatment at well sites, thereby providing a "plug and play" operation. However, the transportation of CNG poses significant challenges, as it necessitates the use of specially designed cylindrical tanks that can withstand the high pressure of the compressed gas.

In the United States, an innovative solution has been implemented to utilize CNG in electrical well intervention operations. A temporary, small-scale plant is established within the field, which takes gas from nearby producer wells, treats it to meet the minimum acceptable parameters, and then compresses it into transportable tanks. This process enables the rapid delivery of the required fuel to operate mobile gas turbine gensets, typically within less than an hour. This solution is particularly well-suited for the Arabian Gulf region, especially for countries that do not produce or only import Liquefied Natural Gas (LNG).

The establishment of small-scale gas treatment and CNG production plants within the field is a feasible solution for the Arabian Gulf region, given the scattered nature of the fields and the associated logistical challenges of transporting gas to well locations. By setting up these plants, operators can reduce the time and cost associated with gas transportation, while also ensuring a reliable and efficient supply of fuel for their mobile gas turbine gensets. This approach can support the region's efforts to reduce its carbon footprint and promote more sustainable energy practices. As the demand for electrical well intervention services will grow in the Arabian Gulf region, the adoption of CNG as a fuel source is likely to play an increasingly important role in supporting the efficient and reliable operation of mobile gas turbine gensets.

Liquefied Petroleum Gas (LPG)

Liquefied Petroleum Gas (LPG) is a fuel source that can be utilized to power mobile gas turbine gensets, albeit it is generally considered a less preferred option compared to other fuels such as Compressed Natural Gas (CNG) or Liquefied Natural Gas (LNG). LPG is a mixture of propane and butane, which are refined from crude oil or natural gas, and is typically stored in a liquid state under pressure. Notably, LPG is widely available in every country in the Arabian Gulf region, making it a readily accessible fuel source.

The use of LPG as a fuel source for mobile gas turbine gensets presents several advantages, including its high energy density, ease of transportation, and relatively low cost. These benefits make LPG an attractive option for powering mobile gas turbine gensets, particularly in regions where other fuel sources may be scarce or expensive. However, LPG also has some significant disadvantages, including its higher emissions of greenhouse gases, and the need for specialized conditioning units at well sites.

Interestingly, LPG is not commonly used as a fuel source for powering mobile gas turbine gensets in countries such as China or the United States, where CNG and LNG are more readily available and preferred due to their cleaner burning properties and lower emissions. However, for the Arabian Gulf region, LPG presents a viable and convenient option for rapidly scaling up electrical well intervention operations, at least until other gas supplies become more widely available. The ease of access to LPG in this region, combined with its relatively low cost and high energy density, make it an attractive interim solution for powering mobile gas turbine gensets and supporting the growth of electrical well intervention operations. As the region continues to develop its energy infrastructure and increase the availability of cleaner-burning fuels, the use of LPG as a fuel source for mobile gas turbine gensets is likely to play an important role in bridging the gap and supporting the transition to more sustainable energy practices.

Required Equipment for Different Fuel Sources (Comparison)

Table 2—Table shows estimations and comparison of most required equipment at well sites for different fuel sources.

EQUIPMENT	LNG/LPG AS FUEL SOURCE	CNG AS FUEL SOURCE	WELLHEAD GAS AS FUEL SOURCE
TRANSPORTED TANK	✓	✓	✗
TEMPORARY PIPELINE	✗	✗	✓
EVAPORATION UNIT	✓	✗	✗
PRESSURE REDUCER	✗	✓	✗
GAS DEHYDRATION UNIT	✗	✗	✓
SOUR GAS TREATMENT	✗	✗	✓
3/4-PHASE SEPARATOR	✗	✗	✓
GAS SURGE TANK	✓	✓	✓

Conclusion

the adoption of electricity-powered equipment for rigless well operations in the Arabian Gulf region is feasible and can significantly reduce Scope 1 and 3 greenhouse gas emissions. While challenges exist, such as reliable electrical energy sources and regional regulations, collaboration and cost-effective solutions can overcome these hurdles. The transition to electricity-powered equipment is expected to continue, driven by the industry's pursuit of efficient and environmentally friendly operations, and the region's commitment to reducing its environmental footprint.

References

- Caines, Jaxon. "US Shale Sector Embraces Rise of the 'G-Fleet'." *J Pet Technol* **77** (2025): 1–6. doi: <https://doi.org/10.2118/0525-0006-JPT>
- Feder, Judy. "E-Fleet —Electric-Powered Fleets Herald a New Future, But It Won't Happen Overnight." *J Pet Technol* **73** (2021): 23–26. doi: <https://doi.org/10.2118/1021-0023-JPT>
- Fu, Dan, Zemlak, Warren, Yeung, Tony, Barclay, Caleb, and Trevor Gorchynski. "Technical, Economic and Environmental Considerations for Selecting Next Generation Hydraulic Fracturing Equipment Technology." Paper presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, USA, October 2022. doi: <https://doi.org/10.2118/210215-MS>